

				Total	B2	[6]
7	(a)	$\lambda = h/p$ or $\lambda = h/mv$ with λ , h and (or mv) p identified			M1 A1	[6] [2]
	(b)	$E = \frac{1}{2} mv^2$ $= p^2/2m$ or $v = \sqrt{(2E/m)}$, <u>hence</u> $\lambda = h/\sqrt{(2mE)}$			C1 M1 A0	[2]
	(c)	$E = qV$ $(0.4 \times 10^{-9})^2 \times 2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19} \times V = (6.63 \times 10^{-34})^2$ $V = 9.4 \text{ V}$ (2 s.f. scores 2/3)			C1 C1 A1	[3]
				Total		[7]
8	(a)	S shown at the peak			B1	[1]